



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
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Smart Textile Manufacturing Quality Inspection and Production Optimization Platform

ARCHITECTURAL DESIGN ASSIGNMENT

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IN

INFORMATION TECHNOLOGY

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1. SYSTEM REQUIREMENT ANALYSIS

1.1 Introduction

The **Smart Textile Manufacturing Quality Inspection and Production Optimization Platform** is designed to improve textile production quality and efficiency. The system combines **AI-based defect detection, IoT sensors, production monitoring, analytics, and centralized reporting**.

The platform reduces dependence on manual inspection and helps factory personnel make better production decisions using real-time and historical data.

1.2 System Objectives

The main objectives are:

- Automate fabric defect detection.
- Classify defects based on type and severity.
- Monitor machines and production in real time.
- Identify possible causes of recurring defects.
- Optimize production using data analysis.
- Generate alerts for abnormal conditions.
- Maintain digital quality and production reports.
- Support different textile manufacturing processes.

1.3 Functional Requirements

The system shall provide:

1. **User Authentication** – Allow authorized users to access the system.
2. **Fabric Inspection** – Capture and process fabric images.
3. **Defect Detection** – Detect fabric defects using AI.
4. **Defect Classification** – Identify defect type, severity, and location.
5. **Production Monitoring** – Monitor machine status, output, and downtime.
6. **Analytics** – Analyze production and defect information.
7. **Alerts** – Notify users when configured limits are exceeded.
8. **Reporting** – Generate quality and production reports.

1.4 Non-Functional Requirements

Scalability

The system should support additional machines, production lines, users, and factories.

Security

Only authorized users should access system functions and production information.

Performance

The system should process fabric images and production data efficiently and provide timely monitoring information.

Reliability

Production and inspection information should be stored accurately without unnecessary data loss.

Availability

The monitoring and important production services should remain available during manufacturing operations.

Maintainability

The system should be divided into manageable components so that individual modules can be modified easily.

1.5 Quality Attributes Influencing Architecture

Quality Attribute	Architectural Requirement
Scalability	Support more machines and users
Security	Protect user and production data
Performance	Fast processing and monitoring
Reliability	Maintain accurate production records
Availability	Keep services accessible
Maintainability	Allow easy modification of components

1.6 Architectural Requirements

The requirements indicate that the system needs:

- Separate processing for AI inspection and normal application operations.
- Real-time communication for machine and sensor data.
- APIs for communication between users and backend services.
- Centralized storage for production and quality information.
- Independent services for reporting, analytics, inspection, and notifications.

Therefore, the architecture must support both **structured application processing and real-time event handling**.

2. SELECTION OF SUITABLE SOFTWARE ARCHITECTURE

2.1 Selected Architecture

The proposed system uses a **Hybrid Architecture** combining:

- Layered Architecture
- Service-Based/Microservice Components
- Event-Driven Communication

The architecture contains the following major parts:

1. Presentation Layer
2. Application Layer / API Gateway
3. Microservices / Service Layer
4. Data Layer
5. Industrial Cameras and IoT Sensors
6. External System Integrations

2.2 Why Hybrid Architecture?

The system has different types of operations. User login, reporting, and dashboard access are normal request-response operations, while machine monitoring, defect detection events, and alerts require real-time processing.

A hybrid architecture is therefore suitable because it combines structured layers with independently operating services and event-based communication.

2.3 Justification

The architecture provides:

- **Scalability** – Individual services can be scaled when required.
- **Maintainability** – Each service has a clear responsibility.
- **Performance** – AI processing can operate separately from normal requests.
- **Real-Time Processing** – Events can trigger immediate actions.
- **Integration** – APIs can connect external systems.
- **Extensibility** – New services can be added later.

2.4 Comparison with Other Architectures

Architecture	Suitability
Layered	Good, but limited for real-time events
Client-Server	Good, but less suitable for distributed services
MVC	Mainly useful for application/UI organization
Microservices	Suitable but more complex
Event-Driven	Good for real-time operations
Hybrid	Most suitable for this project

Therefore, **Hybrid Architecture** is selected because it provides a balance between scalability, real-time processing, maintainability, and complexity.

3. SOFTWARE ARCHITECTURE DIAGRAM

3.1 Architecture Diagram

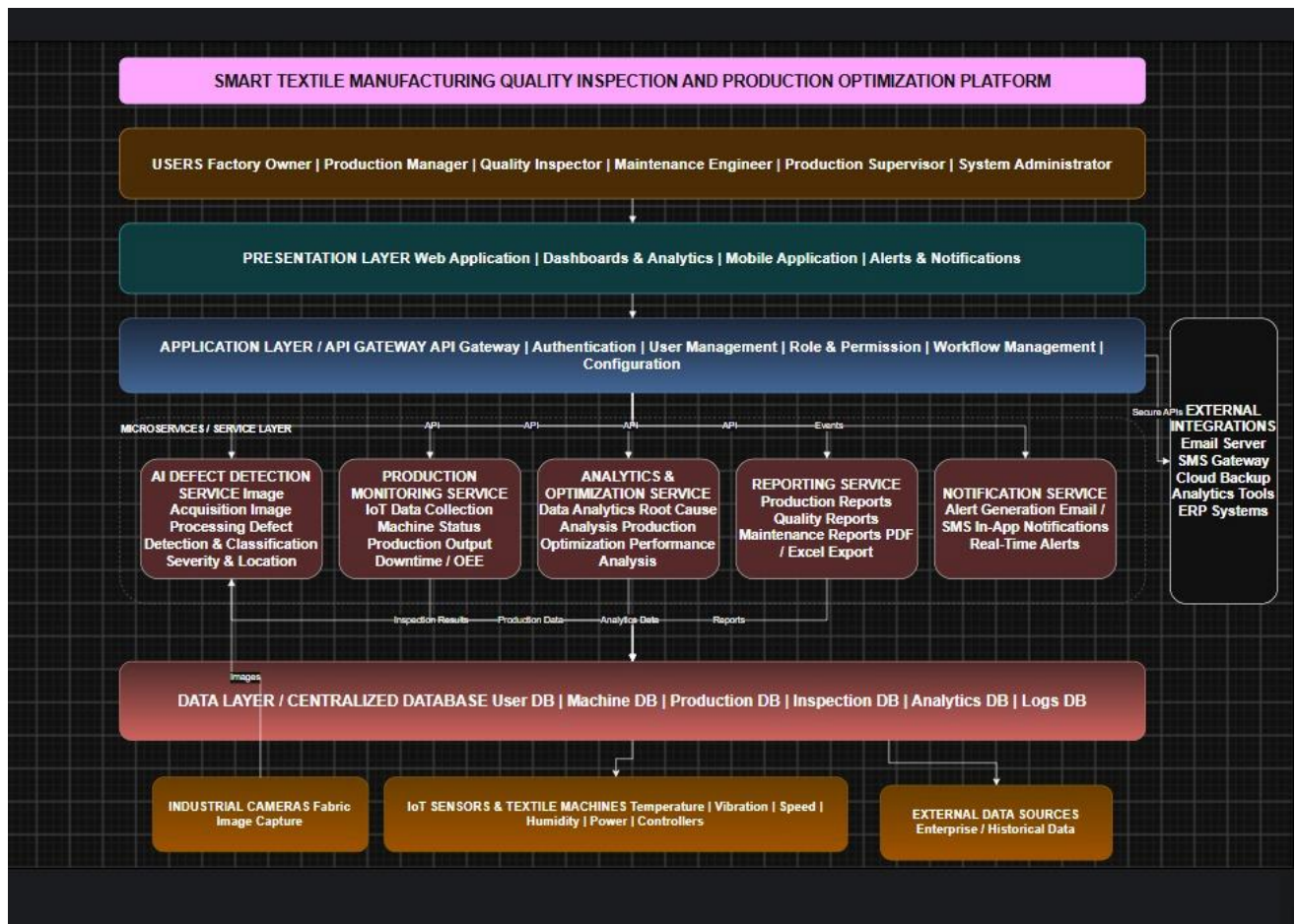


Figure 3.1 – Software Architecture of the Smart Textile Manufacturing Platform

3.2 Diagram Explanation

The architecture begins with users interacting through the **Presentation Layer**. The Presentation Layer contains the web application, dashboards, analytics, mobile access, alerts, and notifications.

Requests are sent to the **Application Layer/API Gateway**, which handles APIs, authentication, user management, roles, permissions, and workflows.

The API Gateway communicates with specialized services in the **Service Layer**, including:

- AI Defect Detection
- Production Monitoring
- Analytics and Optimization
- Reporting
- Notification

The **Data Layer** stores user, machine, production, inspection, analytics, and log information.

Industrial cameras provide fabric images, while IoT sensors and machines provide production information. External integrations can connect services such as email, SMS, cloud backup, analytics tools, and ERP systems.

4. COMPONENTS AND THEIR INTERACTIONS

4.1 Component Responsibilities

Component	Responsibility
Presentation Layer	Provides web/mobile interfaces and dashboards
API Gateway	Handles APIs, authentication and request routing
AI Defect Detection	Detects and classifies fabric defects
Production Monitoring	Monitors machine status, output and downtime
Analytics & Optimization	Performs analysis and production optimization
Reporting Service	Generates production and quality reports
Notification Service	Sends alerts and notifications
Data Layer	Stores system and production information
Industrial Cameras	Capture fabric images
IoT Sensors	Collect machine and environmental data
External Integrations	Connect external services and systems

4.2 Component Communication

The user interacts with the **Presentation Layer**. Requests are sent to the **API Gateway**, which identifies the required service and forwards the request.

The AI service receives fabric images from industrial cameras and produces defect information.

The Production Monitoring Service receives data from machines and IoT sensors and monitors production conditions.

The Analytics Service uses current and historical data for analysis and optimization.

The Reporting Service retrieves stored information and generates reports.

The Notification Service generates alerts when abnormal conditions are detected.

4.3 System Workflow

The overall system workflow is:

Step 1 – Data Collection

Industrial cameras capture fabric images and IoT sensors collect machine data.

Step 2 – Data Transmission

The collected information is sent to the appropriate backend service through APIs or event-based communication.

Step 3 – Processing

The AI service processes fabric images while the monitoring service processes machine information.

Step 4 – Defect Detection

The AI system identifies defects and determines their type, severity, and location.

Step 5 – Data Storage

Inspection and production information is stored in the centralized data layer.

Step 6 – Analysis

The analytics service analyzes production and quality information.

Step 7 – Alert Generation

If a configured threshold is exceeded, the notification service generates an alert.

Step 8 – Reporting

The reporting service generates digital production and quality reports.

Overall Data Flow

Cameras / IoT Sensors → API Gateway / Services → Data Layer → Analytics → Alerts / Reports

This workflow enables continuous monitoring and early identification of production and quality problems.

5. QUALITY ATTRIBUTE ANALYSIS

5.1 Scalability

The architecture allows individual services to be scaled according to workload. Additional machines, production lines, and users can be supported without redesigning the complete system.

5.2 Security

Authentication, role-based access, permissions, API controls, and secure communication help protect users and production information.

5.3 Performance

AI processing, production monitoring, analytics, and reporting are separated into different services. This prevents one heavy operation from unnecessarily affecting other system functions.

5.4 Reliability

Production and inspection information is stored centrally. Separation of services also reduces the impact of failure in one component on unrelated functions.

5.5 Maintainability

Each service has a specific responsibility. Developers can modify the AI, reporting, monitoring, or notification components independently.

5.6 Availability

Important services can operate independently and can be restarted or scaled separately. Future deployment can also use backup and redundancy mechanisms to improve availability.

Summary

The selected architecture addresses all six required quality attributes and provides a suitable foundation for a production-oriented textile manufacturing system.

6. JUSTIFICATION OF ARCHITECTURAL DECISIONS

6.1 API Gateway

An API Gateway provides a common entry point for web and mobile clients. It handles authentication, permissions, and request routing.

Reason: It simplifies communication between clients and backend services.

6.2 Service Separation

AI inspection, production monitoring, analytics, reporting, and notifications are separated into different services.

Reason: Each function has different responsibilities and processing requirements.

6.3 Centralized Data Layer

A centralized data layer stores machine, production, inspection, user, analytics, and log information.

Reason: Root-cause analysis and reporting require information from multiple parts of the system.

6.4 Event-Driven Communication

Events can be used for situations such as:

- High defect rates
- Machine failures
- Production threshold violations
- Maintenance conditions

Reason: Event-based communication supports real-time responses and reduces unnecessary coupling.

6.5 Architectural Trade-Off

The hybrid architecture is more complex than a simple monolithic architecture because it contains multiple services and communication paths.

However, this complexity is justified by the project's requirements for **AI processing, real-time monitoring, automated alerts, scalability, and future integration.**

6.6 Future Enhancements

The selected architecture supports future improvements such as:

- **Predictive Maintenance** using machine sensor data.
- **Advanced AI Models** for improved defect detection.
- **Cloud Deployment** for multiple factories.
- **Mobile Application** for remote monitoring.
- **ERP Integration** for production and inventory management.
- **Additional Textile Processes** such as weaving, knitting, and dyeing.
- **Advanced Analytics** for production forecasting.

6.7 Long-Term Maintainability

The separation of the system into layers and services makes future modifications easier. New services can be introduced without completely redesigning the existing system.

The API-based design also allows future web, mobile, and external applications to communicate with the platform.

6.8 Final Justification

The **Hybrid Architecture** is the most suitable architecture for the Smart Textile Manufacturing Quality Inspection and Production Optimization Platform because it combines structured application layers, specialized services, and event-driven communication.

It supports the major requirements of the system while providing **scalability, security, performance, reliability, maintainability, and availability**.

Therefore, the proposed architecture provides a strong foundation for implementing and expanding the smart textile manufacturing platform.